MATH 468 (Graph Theory): Lecture Notes

Author: Evan Gebhart
Email: ecg7@rice.edu

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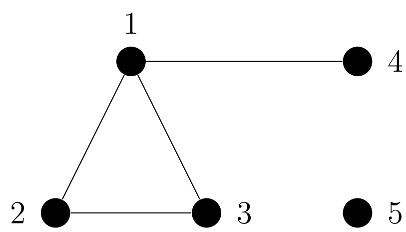
1 Chapter 1: The Basics

1.1 Graphs

Definition (Graph)**.** A *graph* is a pair $G = (V, E)$ of vertices, V (also denoted $V(G)$) and edges E ($E(G)$) such that $E \subseteq V^2$

Definition (Size)**.** The *size* of G, denoted $|G|$, is its order, i.e. the number of vertices.

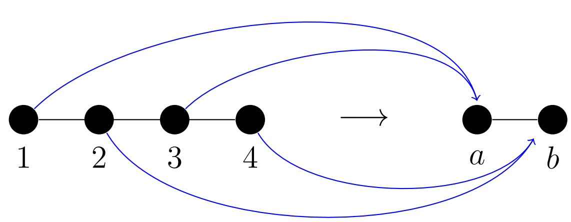


Ex. 1: The graph with vertex set $V = \{1, 2, 3, 4, 5\}$ and edge set $E = \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 4\}\}$

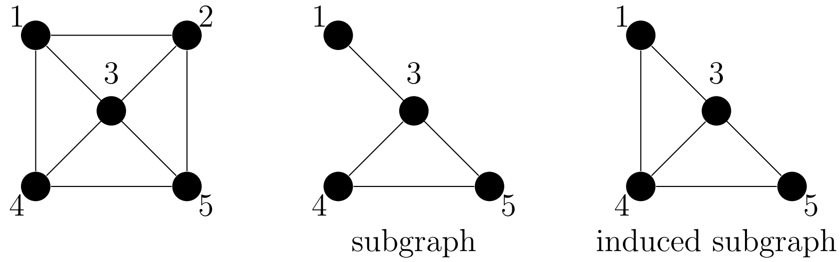
Notation. We typically write an edge, $\{x, y\}$ as xy

Definition. $v \in V$ is incident with edge, e, if $v \in e$. The vertices incident with e are called its *ends*. Two vertices, x, y, are *adjacent* (*neighbors*) if $xy \in E(G)$. Two edges are adjacent if they share a vertex.

Definition (Homeomorphism)**.** Let G, H be graphs. A map $\varphi : V(G) \longrightarrow V(H)$ is a *homeomorphism* if it preserves adjacency, i.e., $xy \in E(G) \implies \varphi(x)\varphi(y) \in E(G)$. If φ is bijective and φ^{-1} is a homeomorphism, then φ is an *isomorphism* and we say G, H are isomorphic (denoted $G \simeq H$). If $\varphi : V(G) \longrightarrow V(G)$, we say that φ is an *automorphism.*



Definition (Subgraph)**.** Let G be a graph. If G' is also a graph and $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$, then G' is a *subgraph* of G. If, additionally, $G' \neq G$, G' is a *proper subgraph* of G. If $U \subseteq V(G)$ and $G' = (U, E(G'))$ where $E(G')$ is exactly the set of edges in G with ends both in U, then G' is an *induced subgraph* of G, denoted $G' = G[U]$

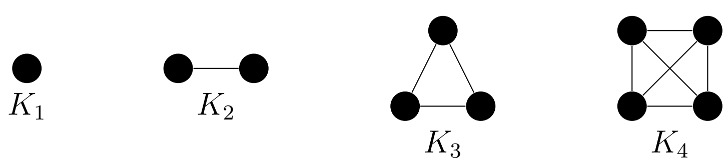


Ex. 2: A graph, G, the subgraph of G defined by $V' = \{1, 3, 4, 5\}$ and $E' = \{13, 34, 35, 45\}$, and the induced subgraph $G[V']$

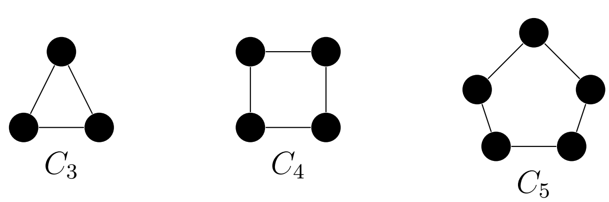
Notation. If $U \subseteq V(G)$, $G - U$ is the graph $G[V(G) \backslash U]$. If $U = \{u\}$ where u is a vertex, then we write $G - u$. Similarly, $G - e$ is the graph with a single edge, e, removed. $G + u$ and $G + e$ are defined similarly.

1.1.1 Some Important Graphs

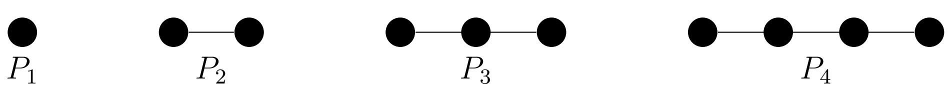
K_n is the *complete* graph on n vertices, that is, the graph obtained by joining every vertex to every other vertex:



C_n is the *cycle* on n vertices (see Section 1.3 for the complete definition of a cycle):



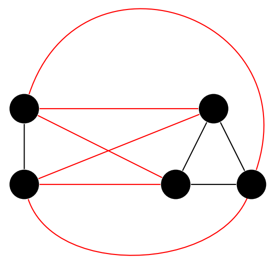
P_n is the *path* on n vertices (see Section 1.3 for the complete definition of a path):



Note that the length of a path is the number of edges, <u>not</u> the number of vertices. Clearly, the length of a P_n path is $n - 1$.

1.1.2 Graph Operations

Definition. $G * H$ is the graph obtained by joining all of the vertices of G to all of the vertices of H



Ex. 3: $K_2 * K_3 = K_5$

Definition (Complement)**.** The *complement* of a graph, denoted $\bar{G}$ is the graph with the vertex set $V(G)$ but exactly the set of edges not in $E(G)$

1.2 The Degree of a Vertex

Notation. Let $v \in V(G)$. The *set of neighbors of* v is denoted $\mathcal{N}_G(v)$ or $\mathcal{N}(v)$. If $U \subseteq V(G)$, the neighbors in $V(G) \backslash U$ of vertices in U is called the *neighbors of* U and denoted $\mathcal{N}(U)$

Definition (Degree)**.** The *degree* $d_G(v) = d(v)$ of a vertex, v, is the number of edges at v. We define $\delta(G) = \min\{d(v) : v \in V\}$ and $\Delta(G) = \max\{d(v) : v \in V\}$. If all of the vertices in G have degree k, then G is *k–regular*. The *average* degree of a graph, denoted $d(G)$, is given by $d(G) = \frac{1}{|V|}\sum_{v \in V} d(v)$. Clearly $\delta(G) \leq d(G) \leq \Delta(G)$.

Proposition 1.2.1. *The number of vertices in a graph of odd degree is even*

Proof. Consider $\sum_{v \in V} d(v)$. Every edge contributes 2 vertices to this sum. Thus

$$2 \cdot |E| = \sum_{v \in V} d(v) \implies |E| = \frac{1}{2}\sum_{v \in V} d(v) \implies \sum_{v \in V} d(v) \text{ is even}$$

□

Definition. The *ratio of the number of edges to the number of vertices* of a graph $G = (V, E)$, denoted $\varepsilon(G)$, is given by $\varepsilon(G) = |E|/|V|$

Remark. By Proposition 1.2.1

$$|E| = \frac{1}{2}\sum_{v \in V} d(v) = \frac{1}{2}d(G) \cdot |V| \longrightarrow \varepsilon(G) = \frac{1}{2}d(G)$$

Proposition 1.2.2. *Every graph, G, with at least one edge has a subgraph, H, with $\delta(H) > \varepsilon(H) \geq \varepsilon(G)$*

Proof. Idea: delete vertices of small degree until you only have vertices of large degree. One can delete a vertex, v, up to $d(v) = \varepsilon(G)$. This deletion lowers $|V(G)|$ by one and lowers $|E(G)|$ by at most $\varepsilon(G)$, so overall $\varepsilon(H)$ does not decrease. Construct a sequence

$$G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \dots$$

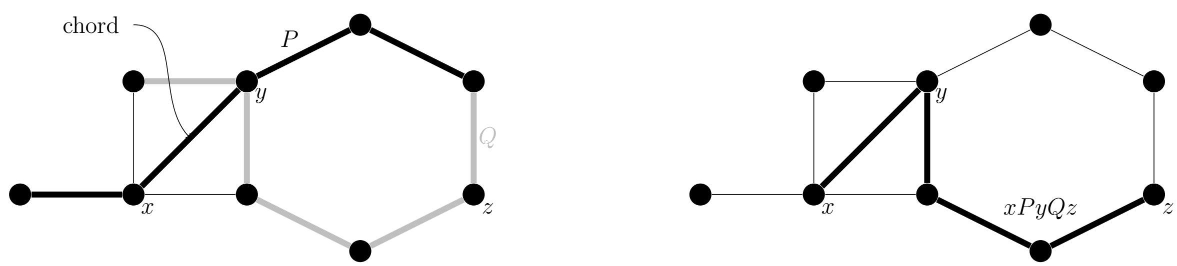
of induced subgraphs of G. If G_i has vertex v_i of degree $d(v_i) \leq \varepsilon(G_i)$, let $G_{i+1} = G_i - v_i$. If not, set $G_i = H$. By choice of G_i, one has $\varepsilon(G_{i+1}) \geq \varepsilon(G_i)$, so $\varepsilon(H) \geq \varepsilon(G)$. Note that $\varepsilon(K_1) = 0$, so the deletion is nontrivial. Since H has no vertex suitable for deletion, $\delta(H) > \varepsilon(H)$ □

1.3 Paths and Cycles

Definition (Path)**.** A *path* is a non-empty graph $P = (V, E)$ of the form $V = \{x_0, x_1, \dots, x_k\}$, with all x_i distinct, and $E = \{x_0x_1, x_1x_2, \dots x_{k-1}x_k\}$. P is said to be an $x_0 - x_k$ path and x_0, x_k are said to be linked by P. The number of edges in a path is its length. We may write $P = x_0x_1 \dots x_k$

Notation. The *subpath* from x_0 to x_i is denoted by $Px_i = x_0x_1 \dots x_i$, while the subpath from x_i to x_k is denoted by $x_iP = x_ix_{i+1} \dots x_k$. The subpath from x_i to x_j is denoted by $x_iPx_j = x_ix_{i+1} \dots x_{j-1}x_j$

Definition (Cycle)**.** A *cycle* is a non-empty graph $C = (V, E)$ of the form $V = \{x_1, \dots, x_k\}$, with all x_i distinct and $|V(G)| > 2$, and $E = \{x_0x_1, \dots x_{k-1}x_k, x_kx_0\}$. The length of a cycle is the number of edges, which is equivalent to the number of vertices. If C has length k, it is a *k–cycle*. An edge that joins 2 vertices on a cycle, but is not on the cycle, is a *chord*. If a cycle has no chords, it is *induced*.



Ex. 4: Left: two paths, P (bold) and Q (gray)
Right: the path $xPyQz$ (bold)

Definition (Girth, Circumference)**.** The minimum length of a cycle is its *girth*, denoted $g(G)$, while the maximum length of a cycle is its *circumference*.

For instance, the graph, G, shown in Example 4 has girth $g(G) = 3$, while the circumference of the graph is 8.

Proposition 1.3.1. *Every graph contains a path of length $\delta(G)$ and a cycle of length $\delta(G)+1$*

Proof. Let $P = x_0x_1 \dots x_k$ be a longest path in G. All neighbors of x_k are in P, so P is at least length $\delta(G)$. Let $i < k$ be minimal such that $x_ix_k \in E(G)$. Then $x_iPx_kx_i$ is a cycle of length at least $\delta(G) + 1$ □

Definition (Distance)**.** The *distance* between two vertices x, y in a graph, G, denoted $d_G(x, y)$ or $d(x, y)$, is the length of the shortest xy path in G. If no path exists, then $d(x, y) \equiv \infty$. The greatest distance between any two vertices in G is the *diameter* of G, denoted $\operatorname{diam}(G)$

Proposition 1.3.2. *Every graph, G, containing a cycle satisfies $g(G) \leq 2 \cdot \operatorname{diam}(G) + 1$*

Proof. Let C be the shortest cycle in G. Assume that $g(G) \geq 2 \cdot \operatorname{diam}(G) + 2$. Then C has two vertices whose distance is at least $\operatorname{diam}(G) + 1$. In G, these vertices have a smaller distance, at most diam(G). The shortest past between these two vertices must not be a subgraph of C. So P contains a C–path xPy. Together with the shorter of the xy paths in C forms a shorter cycle than C, which is a contradiction. □

Definition (Central vertex)**.** A vertex is *central* in G if its greatest distance from any other vertex in G is as small as possible. This distance is the *radius* of G, denoted $\operatorname{rad}(G)$. Formally, $\min \max\{d(x, y) : x, y \in V\}$

Remark. $\operatorname{rad}(G) \leq \operatorname{diam}(G) \leq 2 \cdot \operatorname{rad}(G)$

Proposition 1.3.3. *A graph, G, of radius at most k and max degree $d \geq 3$ has fewer than $\frac{d}{d-2}(d-1)^k$ vertices.*

Proof. Let z be a central vertex in G and let D_i be the set of vertices with distance i from z. Then

$$V(G) = \bigcup_{i=0}^{k} D_i$$

Clearly $|D_0| = 1$ and $|D_1| \leq d$. For $i \geq 1$, one has that $|D_{i+1}| \leq (d-1)|D_i|$ because every vertex in D_{i+1} is a neighbor of a vertex in D_i and each vertex in D_i has at most $d - 1$ neighbors in D_{i+1} (since it has a neighbor in D_{i-1}. Thus, $|D_{i+1}| \leq d(d-1)^i$ for all $i < k$ by induction, giving,

$$|G| \leq 1 + d\sum_{i=0}^{k-1}(d-1)^i = 1 + \frac{d}{d-2}\left((d-1)^k + 1\right) < \frac{d}{d-2}(d-1)^k$$

□

1.4 Connectivity

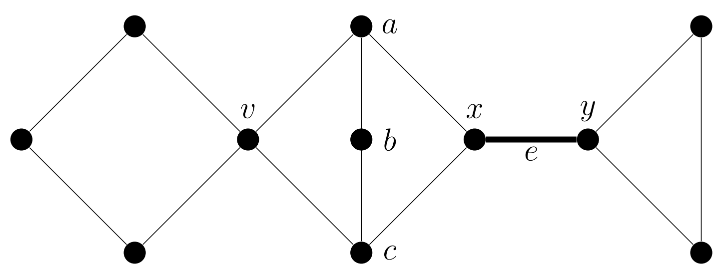
Definition. A graph is called *connected* if it is nonempty and any two vertices are connected by a path

Proposition 1.4.1. *The vertices of a connected graph, G, can be enumerated $v_1, \ldots, v_n$ such that $G_i := G[v_1, \ldots, v_n]$ is connected for all i*

Proof. Pick any vertex v_1. Assume inductively that $v_1, \ldots, v_i$ have been chosen and $G_i = G[v_1, \ldots, v_n]$ is connected. Pick a vertex $v \in G \backslash G_i$. Since G is connected, there exists a $v - v_i$ path P. Change v_{i+1} to be the last vertex of P not in G_i. v_{i+1} has a neighbor in G and thus G_{i+1} is connected. □

Definition (Component)**.** A maximally connected subgraph of G is a *component* of G. The empty graph is not connected and therefore has no components

Definition. Let $A, B \subseteq V(G)$. A set $X \subseteq V(G) \cup E(G)$ **separates** A and B if every $A - B$ path contains a vertex or edge in X. A vertex that separates two other vertices of the same component is called a **cut vertex** and an edge separating its ends is called a **bridge**



Ex. 5: $X = \{a, b, c\}$ separates $\{v\}, \{x, y\}$, v is a cut vertex and the edge e is a bridge

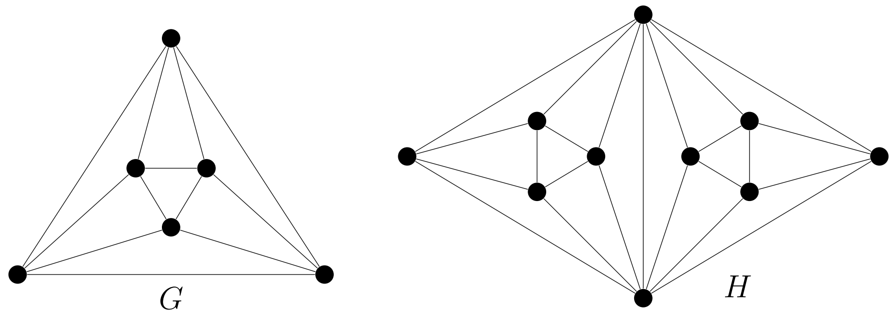
Definition (Connectivity)**.** A graph, G, is called *k-connected* if $|G| > k$ and $G - X$ is connected for all $X \subseteq V(G)$ and $|X| < k$. In other words, you must remove at least k vertices to disconnect G. The greatest $k \in \mathbb{Z}$ such that G is k-connected is called its *connectivity*, denoted $\kappa(G)$

Note. All graphs are 0-connected, all graphs that are connected are 1-connected

Definition. If $|G| > 1$ and $G - F$ is connected for every $F \subseteq E(G)$ of fewer than l edges, G is *l-edge-connected*. The greatest $l \in \mathbb{Z}$ such that G is l-edge-connected is its *edge connectivity*, denoted $\lambda(G)$

Proposition 1.4.2. *If G is nontrivial, $\kappa(G) \leq \lambda(G) \leq \delta(G)$*

Proof. ($\kappa(G) \leq \delta(G)$) To see $\kappa(G) \leq \delta(G)$, let $v \in V(G)$ with degree $\delta(G)$. Remove all the neighbors of v. Then G is disconnected □



Ex. 6: Left: the *octahedron* G with $\kappa(G) = \lambda(G) = 4$
Right: A graph H with $\lambda(H) = 4$, but $\kappa(G) = 2$

Theorem 1.4.3 (Mader, 1972)**.** *Let $0 \neq k \in \mathbb{N}$. Every graph, G, with $d(G) \geq 4k$ has a $(k+1)$–connected subgraph, H, such that $\varepsilon(H) > \varepsilon(G) - k$*

Proof. Recall that $\varepsilon(G) = \frac{1}{2}d(G) \geq 2k$. Let G' be the subgraphs that have the following property

$$|G'| \geq 2k \text{ and } |E(G')| > \varepsilon(G)(|G'| - k) \tag{$\star$}$$

Note that G is one such graph. Let H be such a graph of smallest order. Assume by way of contradiction that G' has order $2k$. Therefore

$$|G'| = 2k \implies |E(G')| > \varepsilon(G)k \geq 2k^2 > \binom{|V(G)|}{2}$$

which is a contradiction. So $|G'| > 2k$. The minimality of H means that $\delta(H) > \varepsilon(G)$ (otherwise we could delete a vertex of degree at most $\varepsilon(G)$ and obtain a smaller graph satisfying ($\star$). Thus $|V(H)| \geq \varepsilon(G)$. Therefore

$$|E(G)| > \varepsilon(G)|V(H)| - \varepsilon(G)k \implies \varepsilon(H) > \varepsilon(G) - \frac{\varepsilon(G)}{|V(H)|}k$$

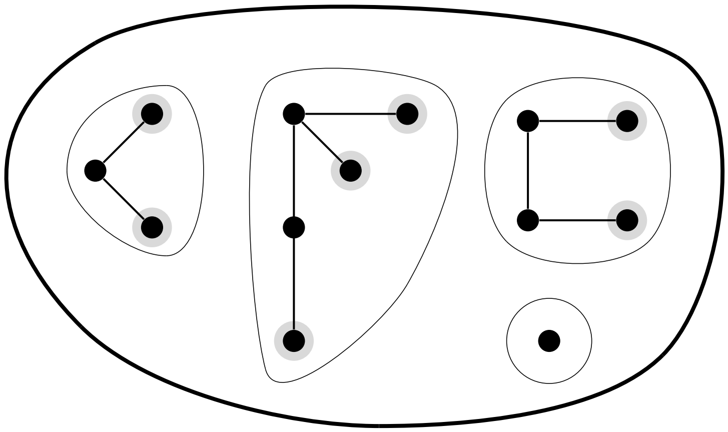
Note that $\frac{\varepsilon(G)}{|V(H)|} < 1$ thus it follows $\varepsilon(H) > \varepsilon(G) - k$. Remains to show that H is $(k+1)$–connected. Assume by way of contradiction it is not. Then there exists $U_1, U_2 \subseteq V(H)$ such that $|U_1 \cap U_2| \leq k$ and $H[U_1 \backslash U_2], H[U_2 \backslash U_1]$ are disconnected. Set $H_1 = H[U_1]$ and $H_2 = H[U_2]$. A $v \in U_1 \backslash U_2$ has degree $d(v) \geq \delta(H) \geq \varepsilon(H)$ neighbors in H_1. In particular, $|H_1| \geq \varepsilon(G) \geq 2k$ (same for H_2). So $|U_1 \cup U_2| \geq 4k$. By the minimality of ($\star$), neither H_1 nor H_2 obey ($\star$). So

$$\begin{aligned}|E(H_1)| \leq \varepsilon(G)(|H| - k) \implies |E(H)| &\leq |E(H_1)| + |E(H_2)| \\ &\leq \varepsilon(G)(|V(H_1) + |V(H_2)| - 2k) \\ &\leq \varepsilon(G)(|V(H)| - k)\end{aligned}$$

which is a contradiction. So H is $(k+1)$–connected $\square$

1.5 Trees and Forests

Definition. A *forest* is an acyclic graph. A *tree* is a forest that is connected (thus the components of a forest are trees). A vertex, v, of a tree with $d(v) = 1$ is a *leaf*. Other vertices are called *inner vertices.*



Ex. 7: Leaves (shaded), Trees (circled), and a Forest (bold circle)

Theorem 1.5.1. *The following four assertions are equivalent:*

1. *T is a tree.*
2. *Any two vertices of T are linked by a unique path.*
3. *T is maximally connected, that is, T is connected, but $T - e$ is disconnected for all $e \in E(T)$*
4. *T is minimally acyclic, that is, T contains no cycles, but $T + xy$ is cyclic for any two non-adjacent vertices x, y*

Remark. *Every nontrivial tree has a leaf. Every connected graph contains a spanning tree.*

Corollary 1.5.2. *The vertices of a tree can always be enumerated $v_1, \ldots v_n$ such that v_i has a neighbor in $v_1, \ldots v_{n-1}$ for all v_i with $i \geq 2$.*

Corollary 1.5.3. *A connected graph with n vertices is a tree if and only if it has $n-1$ edges*

Proof. Consider the case of a single vertex. In this case $n = 1$, and there are $0 = n - 1$ edges. Assume inductively that this holds for a tree on $n-1$ vertices $\longrightarrow n-2$ edges. By the ordering of Corollary 1.5.2, we can add an n^{th} vertex to T. This adds one vertex and one edge. The resulting tree therefore has n vertices and $n-1$ edges. □

Corollary 1.5.4. *For a tree, T, and any graph, G, if $\delta(G) \geq |T| - 1$, then $T \subseteq G$.*

Definition (root)**.** Sometimes, we fix a vertex of a tree as special and call it the *root* of a tree. A tree with a fixed root is called a *rooted tree.*

1.6 Bipartite Graphs

Definition. Let $2 \leq r \in \mathbb{Z}$. A graph $G = (V, E)$ is *r–partite* if there exists a partition of V such that every $e \in E$ has ends in different classes. If $r = 2$, we say that G is *bipartite*

1.6.1 Some Important Classes of r–partite Graphs

$K_{n,m}$ is the class of *complete bipartite graphs*

$K_{n_1,\ldots,n_r}$ is the class of *complete multi–partite Graphs* with vertex sets of size $n_1, \ldots, n_r$. If $n_1 = \cdots = n_r = s$, we write K_s^r

$S_n := K_{1,n-1}$ is the *star* on n vertices.

Proposition 1.6.1. *A graph is bipartite if and only if it contains no odd cycles*

Proof. ($\Longrightarrow$) Let G be a bipartite graph with partite sets X, Y. Consider the cycle $C = v_1 \ldots v_k v_1$. Suppose, without loss of generality, that $v_1 \in X$. Since G is bipartite, $v_i \in X$ for all odd i and $v_i \in Y$ for all even i. Since v_1 is adjacent to v_k, $v_k \in Y$ and k is therefore even.

($\Longleftarrow$) Assume $|V(G)| \geq 2$ and G contains no odd cycles. Let $v \in V(G)$ and define

$$X := \{x \in V(G) : d(x, v) \text{ is even}\}$$
$$Y := \{x \in V(G) : d(x, v) \text{ is odd}\}$$

We will show that G is bipartite with partite sets X, Y. Let $a, b \in X$. Assume, by way of contradiction, that a and b are adjacent. If either a or b are v, the edge ab has length one and is therefore odd, contradicting the definition of X. So neither a nor b are v. Let P_a be the shortest path from v to a and P_b be the same for b. One can enumerate the vertices of P_a as $v_0 v_1 \ldots v_{2k}$ where $v_0 = v$ and $v_{2k} = a$. Similarly, $P_b = w_0 w_1 \ldots w_{2l}$ where $v = w_0, b = w_{2l}$. Let v' be the last vertex that P_a, P_b have in common. Denote P'_a, P'_b as the shortest $v' - a$, $v' - b$ paths respectively where $v' = v_i = w_i$ for some i. Since a, b are adjacent, $v_i v_{i+1} \ldots v_{2k} w_{2k} \ldots w_i$ is a cycle of length $(2k - i) + (2l - i) + 1 = 2(k + l - i) + 1$, which is odd. This contradicts the definition of G. Thus, no two vertices in X are adjacent. The same argument holds for Y. Thus, G is bipartite. □

1.8 Euler Tours

Definition (Euler tour)**.** A *walk* of length k is an alternating sequence of vertices and edges $v_0, e_0, v_1, e_1, \ldots, e_{k-1}, v_k$ such that $e_i = v_i v_{i+1}$ for all $i < k$. If $v_0 = v_k$, the walk is *closed.* A closed walk is an *Euler tour* of a graph, G, if it traverses every edge in G exactly once. A graph is *Eulerian* if it contains an Euler tour.

Theorem 1.8.1 (Euler, 1736)**.** *A connected graph is Eulerian if and only if every vertex has even degree*

Proof. ($\Longrightarrow$) A walk must go into a vertex and out of the same vertex for every edge connected to that vertex. Thus, the vertex must have even degree.

($\Longleftarrow$) Induct on the number of edges: consider a closed walk of maximum length. One can show that this must contain all vertices. Thus it is an Euler tour. $\square$

Remark. *An equivalent definition to Eulerian is a graph where the edges can be partitioned into edge-disjoint cycles*

Definition (Trail)**.** A *trail* is a walk where edges can be repeated

2 Chapter 2: Matching, Covering, and Packing

2.1 Matching in Bipartite Graphs

Note. Throughout this section, all graphs $G = (V, E)$ are bipartite with partite sets A, B (that is, $V = A \cup B$)

Definition (Matching)**.** A set M of independent edges (no shared vertices) in a graph is called a *matching*. M is a matching of $U \subseteq V$ if every $u \in U$ is incident with an $e \in M$. The vertices in U are *matched*. If M is a matching and $U = V$, M is a *perfect matching*.

Definition (k-factor)**.** A k-regular spanning subgraph of a graph G is called a *k-factor*.

[EXAMPLE]

Definition (Alternating & Augmenting Paths)**.** A path that starts at an unmatched vertex in A and contains alternating edges in $E \backslash M$ and M is called an *alternating path*. An alternating path that ends in an unmatched vertex is called an *augmenting path*.

Definition (Cover)**.** A set $U \subseteq V$ is called a (vertex) **cover** if all $e \in E$ are incident with a vertex in U

Theorem 2.1.1 (Kőnig, 1931)**.** *The maximum size of a matching in G a bipartite graph is equal to the minimum size of the cover of its edges*

Proof. Let M be a matching in G of maximum size. From every edge in M let us choose one of its ends: its end is in B if some alternating path ends in that vertex, and its end is in A otherwise. We shall prove that the set U of these $|M|$ vertices covers E; since any vertex U cover of E must cover M, there can be none with fewer than $|M|$ vertices, and so the theorem will follow.

Note that if an alternating path P ends in a vertex $b \in B$, then $b \in U$: as M is a largest matching, P is not an augmenting path, so b is matched to some $a \in A$ and was put in U when we considered the edge $ab \in M$ while constructing U. To show that U covers E, let an edge $ab \in E$ be given. If $a \in U$ we are done, so assume that $a \notin U$. To prove $b \in U$, it suffices to show that some alternating path ends in b. If a is unmatched, then ab is such a path. If not, we have $ab' \in M$ for some $b' \in B$. Since $a \notin U$, there exists an alternating path P ending in b'. Depending on whether or not $b \in P$, either Pb or $Pb'ab$ is an alternating path ending in b. □

A necessary and sufficient condition for the existence of a matching is that every subset of A has "enough" neighbors in B. This is also written as $|\mathcal{N}(S)| \geq |S|$ for all $S \subseteq A$. This condition is called the *marriage condition* and leads to the following theorem

Theorem 2.1.2 (Hall, 1935)**.** *A bipartite graph, G, contains a matching of A if and only if $|\mathcal{N}(S)| \geq |S|$ for all $S \subseteq A$*

Proof. Let $a \in A$ be an unmatched vertex. Let $A' \subseteq A$ be the set of vertices reachable by a nontrivial alternating path and let $B' \subseteq B$ be defined similarly. Notice that $|A'| = |B'|$. Define $S = A' \cup \{a\}$. Clearly, $|S| = |A'| + 1$, thus $|\mathcal{N}(S)| \geq |A'| + 1$. Since $|B'| = |A'|$, by the marriage condition, there exists an edge from some vertex $v \in S$ to some $b \in B \backslash B'$. Since $v \in A' \cup \{a\}$, there exists a path P from a to v. So Pvb is alternating from a to b. Furthermore, since $b \in P$, b is unmatched (if it were, then for some $a', b \in M$, $Pvba'$ is alternating, implying $b \in B'$). Since b is unmatched, Pvb is an augmenting path. □

Corollary 2.1.3. *Every k-regular ($k \geq 1$) bipartite graph has a 1 factor*

Proof. Let G be a k-regular bipartite graph. This implies that $|A| = |B|$. It suffices to show that G contains a matching of A. Every $S \subseteq A$ is joined to $\mathcal{N}(S)$ by $k \cdot |S|$ edges, which are among the $k \cdot |\mathcal{N}(S)|$ edges incident with $\mathcal{N}(S)$. Hence, $k \cdot |S| \leq k \cdot |\mathcal{N}(S)|$, so G satisfies the marriage condition. □

Definition (Stable Matching)**.** A family $(\leq_v)_{v \in V}$ of linear orderings $\leq_v$ on $E(v)$ is a *set of preferences* for G. A matching, M, is *stable* if, for all $e \in E \backslash M$, there exists an $f \in M$ such that e and f have a common vertex with $e <_v f$

Theorem 2.1.4 (Gale & Shapely, 1962)**.** *For every set of preferences on G, a bipartite graph, admits a stable matching*

Proof. M is said to be *strictly better* than M' if $M \neq M'$ and, for every vertex b in an edge $f' \in M'$, b is incident with some $f \in M$ such that $f' \leq_b f$. Each vertex can increase in happiness at most $d(b)$ times, so the process terminates. Given a matching M, a vertex a is *acceptable* to b if $ab := e \in E \backslash M$ and any edge $f \in M$ with $b \in f$ satisfies $f <_b e$. a is said to be *happy* with M if a is unmatched or its edge $f \in M$ has $f >_a e$ for all edges ab where a is acceptable to b

The proof follows by construction: take some $a \in A$ that is unmatched, but acceptable to some $b \in B$. If no b exists, we are done. Add to M the $\leq_a$-maximal edge ab such that

a is acceptable to b. Discard from M any other edge at b. Repeat the process until the terminating condition. Every matching is better than the last and keeps all vertices in A happy. The sequence truncates with a matching M such that no unmatched a is acceptable to any other neighbor in B. Thus, every vertex in A is happy with M, so M is stable. □

Corollary 2.1.5 (Petersen, 1891)**.** *Every regular graph of positive even degree has a 2–factor*

Proof. Let G be any $2k$–regular graph with $k \geq 1$. Suppose, without loss of generality, that G is connected. By Theorem 1.8.1, G contains an Euler tour $v_0 e_0 \ldots e_{l-1} v_l$ with $v_0 = v_l$. We replace every vertex $v \in V(G)$ with the vertex pair (v^-, v^+) and every edge $e_i = v_i v_{i+1}$ by the edge $v_i^+ v_{i+1}^-$. The resulting graph G' is bipartite and k–regular. Thus, by Corollary 2.1.3, G' has a 1–factor. Collapsing every vertex pair (v^-, v^+) back into a single vertex v, we turn this 1–factor of G' into a 2–factor of G. □

2.2 Matching in General Graphs

Notation. Given a graph G, denote by $\mathcal{C}_G$ the set of its components and $q(G)$ the number of odd components in G, that is, the number of components of odd order

Notice, if G has a 1–factor, then $q(G - S) \leq |S|$ for ever $S \subseteq V(G)$. This is because for every odd component of $G - S$, there must be a factor edge in S. This fact is known as *Tutte's Condition* and leads to the following theorem

Theorem 2.2.1 (Tutte, 1947)**.** *A graph G has a 1–factor if and only if $q(G - S) \leq |S|$ for all $S \subseteq V(G)$*

Proof. Assume that G has no 1–factor. We will find some S that violates Tutte's condition. Call such a set a *Tutte violator*. We can assume an even number of vertices, otherwise Tutte's condition is violated by $\emptyset$. We can assume G is edge–maximal, i.e., $G + e$ has a 1–factor for any $e \in E(G)$. If S is a Tutte violator, then, for every spanning subgraph of G, S is also a Tutte violator. Every odd component of $G - S$ can be split into more components, at least one of which will be odd. Define

$$S := \{v \in V : d(v) = |V| - 1\}$$

There are two possible cases:

<u>Case 1:</u> All components of $G - S$ are complete. If $q(G - S) \leq |S|$, we could match one vertex in S with each odd component, pairing up all other vertices, creating a 1–factor.

<u>Case 2:</u> There exists a component, K, of $G - S$ with some $xy \in K$ such that $xy \notin E(G)$. Let P be a shortest xy path that is completely contained in K. Let M_1 be a 1 factor of $G + xb$ where $xb \in M_1$ and let M_2 be defined similarly for $G + ac$ ($ac \in M_2$). Define P' a maximal path starting at c with alternating edges in M_1, M_2. Because c is M_2 matched by ca, the 1st edge of P' is **not** in M_2. This implies the 2nd vertex is M_2 matched and so on.

Let v be the last vertex if P. If the last edge of P is in M_1, $v = a$, otherwise continue with an edge from M_2 and define $C := P' + ac$. If the last edge of P' is in M_2, $v \in \{x, b\}$ and define $C := P' + va + ac$. In either case, C is even in $G + ac$. Every other edge is in M_2. Define $M := M_2 \triangle C$, where $\triangle$ denotes the symmetric difference. M is a perfect matching, which is a contradiction. Therefore, S is a Tutte violator. $\square$

Corollary 2.2.2 (Petersen, 1891)**.** *Every bridgeless 3–regular (cubic) graph has a 1–factor*

Proof. We will show that all bridgeless cubic graphs satisfy Tutte's condition. Let C be an odd component of $G - S$ for $S \subseteq V(G)$. Since G is cubic, the degrees of C $\underline{\text{in } G}$ sum to an odd number, but in C they sum to an even number. So G has an odd number of $S - C$ edges. In particular, there are at least 3 edges between S and C since G is bridgeless. So the total number of edges between S and $G - S$ is at least $3 \cdot q(G - S)$. However, there are at most $3 \cdot |S|$ edges. So $3 \cdot q(G - S) \leq 3 \cdot |S| \implies q(G - S) \leq |S|$ $\square$

3 Chapter 3: Connectivity

Definition. A set F of edges is called *cut* in G if there exists a partition $\{V_1, V_2\}$ of V such that $F = E(V_1, V_2)$. Edges in F are said to *cross* this partition, V_1, V_2 are the *sides* of the cut. A minimal nonempty cut in G is called a *bond*.